GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**SIL Classification Report**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-005
Document Revision	R01
Document Status	Issued for Review
Originator	Val Orafu
Security Classification	Restricted
Issue Date	**08-Aug-22**

SIL Classification Report

R01	08-08-22	Issued for Review	V. Orafu	E. Maduabuchi	W. Irem	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	08.08.22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-005

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

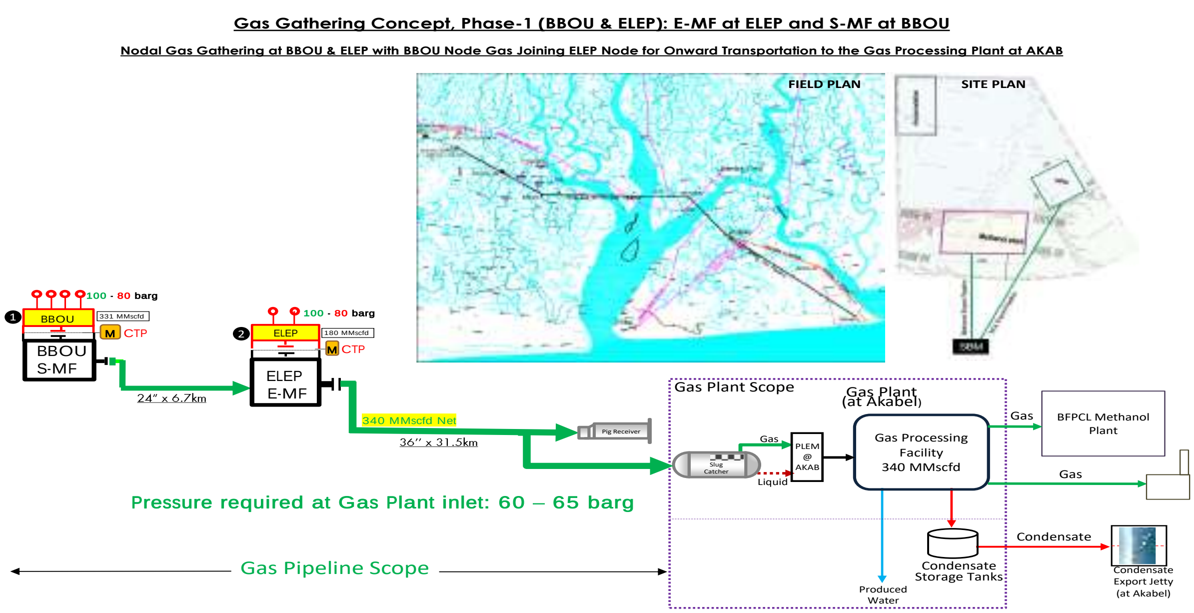

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of the SIL Classification Risk Assessment Study is to:

- Identify the potential safety functions recommended by the Engineering Design HAZOP
- Quantify the level of risk associated with each hazard being protected against
- Determine in each case, the amount of risk reduction necessary to bring the risk to within acceptable (ALARP) limits
- Allocate the risk reduction to the protective layers
- Determine the target Safety Integrity Level (SIL) required of each protective safety function.

The FEED shall fully comply with all specified relevant contractual requirements.

1.3. Scope

The scope of the SIL Classification will cover all SIS within the battery limit of the project for all the manifolds and pipelines

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 80.00.00.15-Gen	Hazard and Operability (HAZOP) Study

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1 Abbreviations

Abbreviation	Meaning
ALARP	As Low As Reasonably Practicable
HAZOP	Hazard and Operability Studies
IEC	International Electrotechnical Commission
IPL	Independent Protection Layer
LOPA	Layer of Protection Analysis
NRV	Non Return Valve
PEFS	Process Engineering Flowscheme
PFD	Probability of Failure on Demand
PFS	Process Flowscheme
PL	Protection Layer
RAM	Risk Assessment Matrix

SIL	Safety Integrity Level
SIS	Safety Instrumented System

3. METHODOLOGY

The SIL classification is a key requirement at this stage of the project, and required by the IEC 61508/61511 standards governing the design of the Safety Instrumented Function (SIF) that is applied for safeguarding all facilities. It is a structured analysis of a SIS, by a multi-discipline team of competent people and chaired by an independent person. The team:

- Identified the risk scenarios from the engineering design HAZOP of the project
- Analysed the initial risk of the scenarios using SPDC RAM (see Appendix 2)
- Conduct Layer of Protection Analysis (LOPA) without taking credit of the SIS, to determine the intermediate risk
- Determine how much additional risk reduction is required to reach the tolerable risk level (SIL determination)

3.1 Rule Set

The following rule sets were adopted for the review

High-high Pressure

Where the occurrence of high pressure is anticipated upon failure of an SIF the following was assumed:

- If the pipe or vessel design pressure is anticipated to be exceeded, leakage from flanges and small-bore fittings can occur.
- If the pipe or vessel test pressure is anticipated to be exceeded, pipe or vessel rupture can occur.

Relief Valves

Where an SIS has parallel mechanical protection offering an equivalent level of protection (e.g. a relief valve), the SIL was established for the combined protection system. It was assumed that a single RV could meet the requirements of at least SIL 1 and the classification of the SIF was reduced by one level, e.g. from SIL 2 to SIL 1.

NRVs

Credit was not taken for single NRV's, because they are never tested and their integrity cannot be verified. However, where two NRV's are installed in series, it was assumed that at least one of them will operate correctly in 9 out of 10 cases of demand, i.e., will reduce the demand rate on the instrumented backflow system by a factor of 10.

Package Equipment Trips

Many of the SISs provide protection for equipment packages such as fired heaters. These equipment packages also have internal protective systems that offer similar levels of protection that will attempt to safely shutdown the equipment should an upset condition occur that goes undetected in the process plant. In the absence of detailed information on the equipment packages and the reliability of their protective systems, no credit has been taken for these systems providing additional protection. Once detailed information is obtained about the packages, credit can be taken for their protective systems, by either:

- Reducing the target SIL of the SIS by reducing the demand rate, or
- Incorporating the package protective system in the test interval analysis of the SIS

Other Assumptions

Risks & Safety Integrity Level

Safety integrity applies to the Electrical / Electronic / Programmable Electronic (E/E/PE) SIF, other technology safety instrumented systems and external risk reduction facilities and is a measure of the likelihood of those systems satisfactorily achieving the necessary risk reduction. Once the tolerable risk has been set, and the necessary risk reduction estimated, the safety integrity requirements for the SIFs can be allocated in terms of the Probability of Failure on Demand (PFD). The average PFD corresponds to one of the SILs specified in Table 6. "SIL a" indicates that although additional mitigation is required, the necessary level of risk reduction is below the SIL1 range and can be achieved through implementation of other measures.

Table 1: SIL Specified PFD

SIL	PFD
SIL 4	≥ 10-5 to < 10-4
SIL 3	≥ 10-4 to < 10-3
SIL 2	≥ 10-3 to < 10-2
SIL 1	≥ 10-2 to < 10-1
SIL a	≥ 10-1 to < 1

Layer of Protection Analysis (LOPA)

The LOPA methodology follows the process described in the AIChE CCPS LOPA 2001. LOPA considers the specified hazards and documents the initiating causes and the protection layers that prevent or mitigate the hazard. The severity of the consequences and the likelihood of occurrence were then compared with applicable risk tolerability criteria and the required SIL determined. It is important to note that when estimating the severity of the consequences, all Independent Protection Layers (IPLs) (general purpose design, basic process control systems, alarms, human actions and physical barriers, including personal protective equipment) are disregarded.

The LOPA process can, in general, be described as follows:

a) Identify hazards (which can be addressed by the implementation of a SIF) using a suitable Process Hazard Analysis tool (e.g. Hazard and Operability Study - HAZOP);

b) Rank the severity of the consequences of the specified hazard. It is important that existing protection layers are disregarded at this stage. Compare this with the corresponding risk target;

c) Identify initiating events and estimate their frequency using operating experience where applicable, data sources and engineering judgment;

d) Identify Conditional Modifiers/Post-Event Mitigation. E.g. occupancy, probability of ignition and vulnerability;

e) Identify Independent Protection Layers (IPLs), which prevent the hazardous event from occurring

f) Compare the Target Risk Frequency with the likelihood of occurrence (Total Mitigated Event Frequency) to determine the required PFD and corresponding SIL target.

Independent Protection Layer (IPL)

In order for an IPL to be considered valid (in accordance with IEC 61511-3), the following criteria were met:

- **Effectiveness** – an IPL reduces the identified risk by at least a factor of 10;
- **Specificity** – an IPL is designed to prevent or mitigate the consequences of one potentially hazardous event. Multiple causes may lead to the same hazardous event, and therefore multiple event scenarios may initiate action by a PL;
- **Independence** – an IPL is independent of other protection layers if it can be demonstrated that there is no potential for common cause or common mode failure with any other claimed IPL;
- **Dependability** – an IPL can be counted on to do what it was designed to do by addressing both random failures and systematic failures during its design;
- **Auditability** – a protection layer is designed to facilitate regular validation of the protective functions.

3.2 Documentation

The following key documentation was used for the SIL Classification study

- Project HAZOP study report

- SPDC Risk Assessment Matrix
- Equipment reliability data based on IEC 61508 – see Appendix 3

3.3 SIL Classification Agenda

The SIL classification was carried out as a desktop exercise

- SIL Classification Facilitator listed all other independent protection layers (IPL) on the SISs from the HAZOP study
- SIL team carried out a risk assessment ranking of the worst possible consequences
- SIL team carried out a SIL risk assessment based on the reliability data on the initiating events and probability of failure on demand (PFD) of the IPLs
- SIL team determined the SIL level based on SIL Calculation Sheet developed by the Consultant (ESLP Consultants Ltd) based on IEC 61508
- All relevant information were recorded on the worksheet.

4. SIL Classification Results

A total of 6 SISs were SIL-classified, the summary of the classification is as shown below.

Please note: The SIL classification of new SISs that may arise from close out of the HAZOP report needs to be carried out.

SIL	No of SIS
No SIL	4
SIL1	1
SIL 2	0
SIL 3	1
SIL 4	0

Details of the SIL classification result is shown in Table 1 below

S/N	Cause	Consequences	SIS TAG	SIL Level	Remark
BBOU					
1	Fail close of SDV 1001/1002 Inadvertent closure of manual valves of the line	Pressure build up upstream the blockage up the CITHP of SPDC BBOU well (190barg) potentially leading to rupture of the piping and LOPC, resulting into injuries/fatalities from ignited release and environmental pollution from unignited release (Piping is 900lb class)	BBOU PSD1 HH	SIL 3	SIL level is dependent on the SIL level of the overpressure protection in SPDC manifold upstream – SIL 1 was assumed. If this SIL level is SIL 3, BBOU PSD1 will become SIL a
2	Rupture of the pipeline from external impact (excavation)	LOPC resulting into injuries/fatalities from ignited release and environmental pollution from unignited release	BBOU PSD1 LL	SIL 1	
ELEP					

3	Fail close of SDV 1001/1002 Inadvertent closure of manual valves of the line	Pressure build up upstream the blockage up the CITHP of SPDC BBOU well (190barg) potentially leading to rupture of the piping and LOPC, resulting into injuries/fatalities from ignited release and environmental pollution from unignited release (Piping is 900lb class)	ELEP PSD1 HH	SIL a	SIL level is dependent on the SIL level of BBOU PSD 1 which is dependent on the SIL Level of upstream SPDC manifold
4	Rupture of the pipeline from external impact (excavation)	LOPC resulting into injuries/fatalities from ignited release and environmental pollution from unignited release	ELEP PSD1 LL	SIL a	
AKAB					
5	Fail close of SDV 1001/1002 Inadvertent closure of manual valves of the line	Pressure build up upstream the blockage up the CITHP of SPDC BBOU well (190barg) potentially leading to rupture of the piping and LOPC, resulting into injuries/fatalities from ignited release and environmental pollution from unignited release (Piping is 900lb class)	AKAB PSD1 HH	SIL a	SIL level is dependent on the SIL level of BBOU PSD 1 which is dependent on the SIL Level of upstream SPDC manifold
6	Rupture of the pipeline from external impact (excavation)	LOPC resulting into injuries/fatalities from ignited release and environmental pollution from unignited release	AKAB PSD1 LL	SIL a	

5. Discussions, Conclusions & Recommendations

The SIL level of the overpressure protection of the upstream SPDC manifold is critical in the SIL determination of the SIS overpressure protection in the project

The table below provides a summary of discussions and recommendations only where the classification concluded that No SIL is required.

SIS description	SIL No's	Design Recommendation
High pressure protection on the ELEP pipeline system (ELEP PSD1 HH)	3	Maximum consequence of a high pressure on this pipeline system is flange leaks, the piping is also fully rated to the upstream pressure, subject to SPDC confirmation of the upstream pressure. It is recommended that the SIS be retained, even though an alarm function is deemed sufficient
Low pressure protection (spill downstream) on the ELEP pipeline system (ELEP PSD1 LL)	4	Pipeline could be ruptured at the buried sections on the RoW and this may result into significant environmental pollution. Credit has been taken for the PSD1 LL upstream at BBOU. It is recommended that the SIS be retained and possibly upgraded to SIL 1 (same level with upstream BBOU PSD1 LL)
High pressure protection on the AKAB pipeline system (AKAB PSD1 HH)	5	Maximum consequence of a high pressure on this pipeline system is flange leaks, the piping is also fully rated to the upstream pressure, subject to SPDC confirmation of the upstream pressure. It is recommended that the SIS be retained, even though an alarm function is deemed sufficient
Low pressure protection (spill downstream) on the AKAB pipeline system (AKAB PSD1 LL)	6	Pipeline could be ruptured at the buried sections on the RoW and this may result into significant environmental pollution. Credit has been taken for the PSD1 LL upstream at BBOU and ELEP. It is recommended that the SIS be retained and possibly upgraded to SIL 1 (same level with upstream BBOU PSD1 LL and ELEP PSD1 LL)

Appendix 1: SIL Classsification Summary

Scenario Number	Equipment No:	SIF Tag No:	BBOU PSD1 HH	SIL Rating
				SIL 3
	Description	**Probability**	**Frequency (per/year)**	
Impact Description				
Risk Tolerance Criteria			1.00E-05	
Initiating Event Frequency	SDV failure		1.00E-01	
Enabling Event	% of the time in operation	1		
Conditional Modifiers	Ignition Probability	1.00E+00		
	Probability of personnel in affected area	1		
	Others	1		
Frequency of Unmitigated Consequence			1.00E-01	
Independent Protection Layers (IPL)				
Basis Process Control System (BPCS)		1.00E+00		
Human Intervention		1.00E+00		
SIF	SIF Upstream by SPDC	0.1	SIL 1 assumed	
Additional Mitigation				
PSVs	PSV-1001	1.00E+00	Fire case	
Other IPLs 1		1.00E+00		
Other IPLs 2		1.00E+00		
Total PFD for all IPLs		1.00E-01		
Frequency of Unmitigated Consequence			1.00E-02	
Risk Reduction Factor			1.00E+03	

Scenario Number	Equipment No:	SIF Tag No:	**BBOU PSD1 LL**	SIL Rating
				SIL 1
	Description	**Probability**	**Frequency (per/year)**	
Impact Description				
Risk Tolerance Criteria			1.00E-05	
Initiating Event Frequency	External Impact (Excavation)		9.00E-04	
Enabling Event	% of the time in operation	1		
Conditional Modifiers	Ignition Probability	1.00E+00		
	Probability of personnel in affected area	1		
	Others	1		
Frequency of Unmitigated Consequence			9.00E-04	
Independent Protection Layers (IPL)				
Basis Process Control System (BPCS)		1.00E+00		
Human Intervention		1.00E+00		
SIF	SIF Upstream by SPDC	1.00E+00		
Additional Mitigation				
PSVs		1.00E+00		
Other IPLs 1		1.00E+00		
Other IPLs 2		1.00E+00		
Total PFD for all IPLs		1.00E+00		
Frequency of Unmitigated Consequence			9.00E-04	
Risk Reduction Factor			9.00E+01	

Scenario Number	Equipment No:	SIF Tag No:	**ELEP PSD1 HH**	SIL Rating
				SIL a
	Description	**Probability**	**Frequency (per/year)**	
Impact Description				
Risk Tolerance Criteria			1.00E-05	
Initiating Event Frequency	SDV failure		1.00E-01	
Enabling Event	% of the time in operation	1		
Conditional Modifiers	Ignition Probability	1.00E+00		
	Probability of personnel in affected area	1		
	Others	1		
Frequency of Unmitigated Consequence			1.00E-01	
Independent Protection Layers (IPL)				
Basis Process Control System (BPCS)		1.00E+00		
Human Intervention		1.00E+00		
SIF	SIF Upstream by SPDC	0.1	SIL 1 assumed	
Additional Mitigation				
PSVs	PSV-1001	1.00E+00	Fire case	
Other IPLs 1	BBOU PSD1 HH	1.00E-03	SIL 3	
Other IPLs 2		1.00E+00		
Total PFD for all IPLs		1.00E-04		
Frequency of Unmitigated Consequence			1.00E-05	
Risk Reduction Factor			1.00E+00	

Scenario Number	Equipment No:	SIF Tag No:	**ELEP PSD1 LL**	SIL Rating
				SIL a
	Description	**Probability**	**Frequency (per/year)**	
Impact Description				
Risk Tolerance Criteria			1.00E-05	
Initiating Event Frequency	External Impact (Excavation)		9.00E-04	
Enabling Event	% of the time in operation	1		
Conditional Modifiers	Ignition Probability	1.00E+00		
	Probability of personnel in affected area	1		
	Others	1		
Frequency of Unmitigated Consequence			9.00E-04	
Independent Protection Layers (IPL)				
Basis Process Control System (BPCS)		1.00E+00		
Human Intervention		1.00E+00		
SIF	BBOU PSD1 LL	1.00E-01	SIL 1	
Additional Mitigation				
PSVs		1.00E+00		
Other IPLs 1		1.00E+00		
Other IPLs 2		1.00E+00		
Total PFD for all IPLs		1.00E-01		
Frequency of Unmitigated Consequence			9.00E-05	
Risk Reduction Factor			9.00E+00	

Scenario Number	Equipment No:	SIF Tag No:	**AKAB PSD1 HH**	SIL Rating
				SIL a
	Description	**Probability**	**Frequency (per/year)**	
Impact Description				
Risk Tolerance Criteria			1.00E-05	
Initiating Event Frequency	SDV failure		1.00E-01	
Enabling Event	% of the time in operation	1		
Conditional Modifiers	Ignition Probability	1.00E+00		
	Probability of personnel in affected area	1		
	Others	1		
Frequency of Unmitigated Consequence			1.00E-01	
Independent Protection Layers (IPL)				
Basis Process Control System (BPCS)		1.00E+00		
Human Intervention		1.00E+00		
SIF	SIF Upstream by SPDC	0.1	SIL 1 assumed	
Additional Mitigation				
PSVs	PSV-1001	1.00E+00	Fire case	
Other IPLs 1	SIF Upstream by SPDC	1.00E-01		
Other IPLs 2	BBOU PSD1	1.00E-01		
Other IPLs 3	ELEP PSD1	1.00E-03		
Total PFD for all IPLs		1.00E-05		
Frequency of Unmitigated Consequence			1.00E-06	
Risk Reduction Factor			1.00E-01	

Scenario Number	Equipment No:	SIF Tag No:	**AKAB PSD1 LL**	SIL Rating
				SIL a
	Description	**Probability**	**Frequency (per/year)**	
Impact Description				
Risk Tolerance Criteria			1.00E-05	
Initiating Event Frequency	External Impact (Excavation)		9.00E-04	
Enabling Event	% of the time in operation	1		
Conditional Modifiers	Ignition Probability	1.00E+00		
	Probability of personnel in affected area	1		
	Others	1		
Frequency of Unmitigated Consequence			9.00E-04	
Independent Protection Layers (IPL)				
Basis Process Control System (BPCS)		1.00E+00		
Human Intervention		1.00E+00		
SIF	BBOU PSD1 LL	1.00E-01	SIL 1	
Additional Mitigation				
PSVs		1.00E+00		
Other IPLs 1		1.00E+00		
Other IPLs 2		1.00E+00		
Total PFD for all IPLs		1.00E-01		
Frequency of Unmitigated Consequence			9.00E-05	
Risk Reduction Factor			9.00E+00	

Appendix 2: SPDC Risk Assessment Matrix

SEVERITY	CONSEQUENCES				INCREASING LIKELIHOOD				
					A	B	C	D	E
	People	Assets	Community	Environment	Never heard of in the industry	Heard of in the industry	Has happened in the Organization or more than once in the Industry	Has happened at the Location or more than once per year in the Organization	Has happened more than once per year at the location
0	No injury or health effect	No damage	No impact	No effect					
1	Slight injury or health effect	Slight damage	Slight impact	Slight effect					
2	Minor injury or health effect	Minor damage	Minor impact	Minor effect					
3	Major injury or health effect	Moderate damage	Moderate impact	Moderate effect					
4	PTD or up to 3 fatalities	Major damage	Major impact	Major effect					
5	More than 3 fatalities	Massive damage	Massive impact	Massive effect					

Appendix 3: SIL Classification Data

3.1: Initiating Events Data for SIL

Initiating Event	**Frequency Rate** (to be used for SIL Allocation) **(per year)**
Pressure Vessel Residual Failure	1×10^{-6}
Piping Leak (10% section-100m)	1×10^{-3}
Atmospheric Tank Failure	1×10^{-3}
Third Party Intervention (external impact)	1×10^{-2}
Safety Valves open spuriously	1×10^{-2}
Cooling Water Failure	1×10^{-2}
Unloading / Loading hose failure	1×10^{-2}
BPCS (or DCS) instrument loop failure	1×10^{-1}
Control valve failure	1×10^{-1}
Human error – trained, without adequate operating procedure, under stress	1×10^{-1}
Human error – trained, with adequate operating procedure	1×10^{-2}
Manual Valve - catastrophic failure	1×10^{-2}
Check Valve - catastrophic failure (no Inspection, testing and preventive maintenance (ITPM) program)	1×10^{-1}
Motor operated valves-spurious operation	1×10^{-2}

3.2.1: Protection Layer Data

Protection Layer Database	PFD_{avg}
Inherently safe design	1×10^{-2}
Open vent (no valve) - for over-pressure protection	1×10^{-2}
PLC system interlock (separate from BPCS)	1×10^{-1}
Check valve (for reverse flow)	1×10^{-1}
BPCS Instrument loop	1×10^{-1}
Administrative controls (such as CSO/CSC, operational check-list)	1×10^{-1}
Flame scanner (2oo2)	6×10^{-2}
Control valve failure	1×10^{-1}
Human action with 10min response time	1×10^{-1}
Human action with 40min response time	1×10^{-2}
Manual Valve - catastrophic failure	1×10^{-3}

3.2.2: Protection Layer Data

Protection Layer Database	PFD_{avg}
Check Valve - catastrophic failure	$1x10^{-2}$
Motor operated valves-spurious operation	$1x10^{-2}$
Rupture disc	$1x10^{-2}$
SIS SIL-1 safety loop	$1x10^{-1}$
SIS SIL-2 safety loop	$1x10^{-2}$
SIS SIL-3 safety loop	$1x10^{-3}$
Vacuum breaker	$1x10^{-2}$
Pressure relief valve	$1x10^{-2}$
Dike	$1x10^{-2}$
Underground Drainage System	$1x10^{-2}$
Fire Proofing	$1x10^{-2}$
Blast-wall/Bunker	$1x10^{-3}$
Flame/Detonation Arrestor	$1x10^{-2}$